Design Brief – Review 2

0. Header

Project code	FA26SE153
Project name	HR Connect: AI-Powered Recruitment & Affiliate Headhunting Platform
Team	Trương Hoàng Quí (Leader), Khúc Ngọc Sơn, Cao Hữu Trí, Lê Thị Trà Mi, Nguyễn Văn Sang
Supervisor	Phạm Minh Trí (tripm14@fpt.edu.vn)
Repository link (tag review-2)	https://github.com/hoangqui794/hr-connect (tag review-2)
Version	1.0
Date	23/10/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Actors	Admin/HR (one role)	Internal HR + Platform Admin	Separation of duties
Post-hire	Probation, commission at probation pass	30-day warranty, commission after warranty and Client payment	Partner rule 17/10
Job lifecycle	Draft/Active/Paused/Closed	Adds review, agreement, payment and rejected states	Fee, deposit, agreement before go-live
Agreement	—	OTP-signed service agreement (Headhunt, Sourcing)	Partner requirement
Submission	Duplicate check	Plus e-mail consent and source disputes	Personal data decree
AI	Keyword pre-filter	Per-requirement LLM assessment with verified evidence	Explainability
NFR / tech	AI ≤ 5 s; Material UI, MinIO	Async AI ≤ 60 s p95; Ant Design, R2, Render/Vercel	Iteration 1 findings

1.2 Out of Scope

Item	Reason
Native mobile application	Responsive web covers all roles within the capstone time.
Online payment gateway	Fees, deposits and payouts are paid outside the system; Admin records the reference.
Probation tracking	Replaced by the warranty rule (BM03 change 2).
Internal HR submitting CVs for candidates	Team decision 16/10/2026; submissions come only from candidates and affiliates.

2. System at a Glance

Small recruitment agencies run headhunting, CV sourcing and CV application services through spreadsheets and chat groups, which causes duplicated candidates, commission disputes with freelance recruiters and slow manual CV reading. HR Connect is one web platform for Client companies, Candidates, Affiliates (freelance headhunters), Internal HR and Platform Admin. It manages the whole pipeline from job approval and signed

service agreement, through consented candidate referral with duplicate blocking and first-source attribution, AI scoring with evidence, interview and offer, to warranty, service fees and affiliate commission payouts. Architecture: React web app, ASP.NET Core 8 API with PostgreSQL, and a separate FastAPI AI service (SDD Report 4, §1.1).

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Backend structure	Layered MVC; Clean Architecture with vertical feature slices	Clean Architecture + MediatR per feature: rules testable without the web layer, members do not collide.	Trương Hoàng Quí	Report 4 §1.1, §1.2
D2	AI scoring integration	In-process scoring; synchronous HTTP call; separate service with queue and callback	Separate FastAPI service, async queue + callback, SBERT + LLM with verified evidence: slow LLM does not block users.	Trương Hoàng Quí	F-09, Report 4 §3.3
D3	Access by service type	Role checks in code; per-job visibility flag; configurable role × service-type matrix	ServiceTypeAllowedRole (can_view / can_submit) configured by Admin: rules change without code changes.	Lê Thị Trà Mi	F-01, F-05, sm-job
D4	Concurrent edits	Last write wins; pessimistic locks; optimistic ConcurrencyToken	Optimistic ConcurrencyToken + HTTP 409 on stale writes: safe for several HR/Client users without long locks.	Khúc Ngọc Sơn	F-10, F-13, Report 4 §2
D5	Commission and fee ledger	Compute on the fly; store final amount only; store snapshots with state machine	Commission snapshots + state machine gated by warranty and Client payment.	Nguyễn Văn Sang	F-15..F-17, sm-commission

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	LLM provider outage or rate limit	Applications stay unscored; HR cannot rank CVs	Fallback models/providers, retry with back-off, FAILED status with manual retry; test DS-06.	Trương Hoàng Quí
R2	AI service hosted on a personal machine through a tunnel	Staging demo fails if the machine is off	Health check before demo; move AI service to a cloud host before Defense 1.1.	Trương Hoàng Quí
R3	Time-based rules (warranty, consent expiry, offer expiry) hard to demo and test	Demo scenarios cannot finish in the session	Durations configurable via settings; seeded data near deadlines; unit tests for the workers.	Nguyễn Văn Sang

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	Company directory with LinkedIn industry filter has API but no public page	Registration 3.2.c Reporting; FE	Build the public directory page	Nguyễn Văn Sang	Before Defense 1.1

ID	Gap	Location	Plan	Owner	Due
G2	NFR N1 load test with 50,000 profiles not yet run	Registration 3.2.d	Seed data script and k6 test on staging	Trương Hoàng Quí	Before Defense 1.1
G3	Self-review (BM02) not yet run on tag review-2	BM02	Tag repo, run prompt, list High items	Trương Hoàng Quí	Before submission
G4	State machines need owner check	Report 4 §4	Owners verify vs handlers	Workflow owners	Before Review 2

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Create and submit job	Client	No MUST_HAVE requirement → submit blocked; unverified company → blocked; stale token → 409.	TC-F01 ×3	Done
F-02	Review job	Internal HR	Reject needs a reason code; approve sends HEADHUNT/SOURCING to PENDING_AGREEMENT.	TC-F02 ×3	Done
F-03	Service agreement OTP signing	Internal HR, Client	Only the primary contact signs; wrong/expired OTP rejected; signed PDF stored.	TC-F03 ×3	Done
F-04	Record posting fee / deposit	Platform Admin	No reference → rejected; Internal HR → 403; job becomes ACTIVE.	TC-F04 ×3	Done
F-05	Candidate applies to job	Candidate	Second apply to same job or apply to PAUSED job → blocked; AI scoring queued.	TC-F05 ×3	Done
F-06	Affiliate referral with consent	Affiliate, Candidate	Application created only after consent; resend inside cool-down → blocked.	TC-F06 ×3	Done
F-07	Duplicate blocking and attribution	System, Affiliate	Same candidate, same job → BLOCKED_DUPLICATE, first attribution kept.	TC-F07 ×3	Done
F-08	Source dispute	Affiliate, Platform Admin	Only the blocked affiliate can open; other name hidden; UPHELD moves attribution.	TC-F08 ×3	Done
F-09	AI scoring with tier and evidence	System	Score 0–100, configured tier, evidence found in CV; failure → FAILED + retry.	TC-F09 ×3	Done
F-10	HR screening	Internal HR	Reject needs reason; CV_APPLICATION → 403; concurrent change → 409.	TC-F10 ×3	Done
F-11	Sourcing CV delivery and Client decision	Internal HR, Client	Beyond Sourcing target → 409; reject needs reason; overdue → auto-accepted.	TC-F11 ×3	Done
F-12	Interview scheduling and result	Client, Candidate	Only company members schedule; candidate notified in real time; HR → 403.	TC-F12 ×3	Done
F-13	Offer	Client, Candidate	Headhunt offer must be gross monthly; withdraw needs reason; accepted cannot be withdrawn.	TC-F13 ×3	Done
F-14	Confirm start work	Client	Future date or date before acceptance → rejected; valid date creates placement, fee, warranty.	TC-F14 ×3	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-15	Warranty and resignation	Client, Internal HR	Report → commission ON_HOLD; HR confirm → warranty VOIDED, replacement slot (no new fee).	TC-F15 ×3	Done
F-16	Service fee payment	Platform Admin	Fee = multiplier × gross monthly salary; NOT_STARTED → no fee; payment needs reference.	TC-F16 ×3	Done
F-17	Commission approval and payout	Platform Admin, Affiliate	Blocked while fee unpaid or dispute open; failed payout retried; affiliate sees history.	TC-F17 ×3	Done

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
N1	Performance	p95 response time of public job search with filters	≤ 2 s (5k jobs, 50k profiles)	k6 on staging, 50 users, 5 min
N2	Performance (AI)	Time from application created to AI result COMPLETED	95% ≤ 60 s	AiMatchResult timestamps, 50 applications
N3	Security	Wrong-role requests rejected	100% return 401/403	Authorization tests in CI
N4	Reliability	Exact e-mail/phone duplicates blocked	100% blocked	Duplicate test set in CI
N5	Auditability	Attribution/commission changes audited	100%	Integration tests on AuditLog
N6	Real-time	Delay from commit to in-app notification	≤ 3 s	Playwright timing of bell update

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Trương Hoàng Quí	Leader	AI service; architecture, CI/deploy	WF-03	D1, D2
Cao Hữu Trí	Member	Submission, consent, attribution; disputes	WF-02	—
Khúc Ngọc Sơn	Member	Interview, offer, placement	WF-04	D4
Lê Thị Trà Mi	Member	Job, service agreement	WF-01	D3
Nguyễn Văn Sang	Member	Finance (fee, warranty, commission); frontend/UX	WF-05	D5

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Create, approve and publish a job	Client, Internal HR, Platform Admin	Client submits a job	Job ACTIVE and visible to allowed roles	F-01, F-02, F-03, F-04	Job: DRAFT → PENDING_REVIEW → PENDING_AGREEMENT → PENDING_PAYMENT → ACTIVE	Lê Thị Trà Mi
WF-02	Refer a candidate, consent and attribution	Affiliate, Candidate, Platform Admin	Affiliate submits a candidate	Application created and attributed to the first valid source	F-05, F-06, F-07, F-08	Submission: PENDING_CONSENT → ACCEPTED; Consent: PENDING → CONFIRMED	Cao Hữu Trí
WF-03	AI matching and screening	Internal HR, Client (AI service)	Application created	Ranked, explained shortlist delivered to the Client	F-09, F-10, F-11	AiMatchResult: PENDING → PROCESSING → COMPLETED; Application: SUBMITTED → SCREENING → SHORTLISTED	Trương Hoàng Quí
WF-04	Interview, offer and start work	Client, Candidate	Client schedules an interview	Candidate placed	F-12, F-13, F-14	Application: SHORTLISTED → INTERVIEW → OFFER_PENDING → OFFER_ACCEPTED → PLACED; Offer: DRAFT → SENT → ACCEPTED	Khúc Ngọc Sơn
WF-05	Warranty, service fee, commission and payout	Client, Internal HR, Platform Admin, Affiliate	Placement started	Fee paid and commission paid to the affiliate	F-15, F-16, F-17	Warranty: ACTIVE → PASSED; Commission: IN_WARRANTY → EARNED → PAYABLE → PAID	Nguyễn Văn Sang

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	Verified company, HEADHUNT_COD job with 2 MUST_HAVE requirements	Submit → HR approves → HR sends agreement → Client signs with OTP	Job ACTIVE; signed PDF downloadable	Lê Thị Trà Mi
DS-02	WF-01	Exception	Agreement sent to the Client	Client enters a wrong OTP	Signing rejected; job stays PENDING_AGREEMENT	Lê Thị Trà Mi
DS-03	WF-02	Main	Active HEADHUNT_COD job, approved affiliate, new candidate e-mail	Affiliate refers → candidate accepts consent	Application SUBMITTED, attribution to the affiliate, affiliate notified	Cao Hữu Trí

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04	WF-02	Exception	Candidate already referred to the same job	Second affiliate refers the same candidate	BLOCKED_DUPLICATE; first attribution kept	Cao Hữu Trí
DS-05	WF-03	Main	Application with a text-layer CV, AI service online	AI scores → HR shortlists with reason	Score, tier and evidence shown; Client sees masked CV	Trương Hoàng Quí
DS-06	WF-03	Exception	AI service stopped	Application created; HR retries after the service is back	Result FAILED with message, then COMPLETED after retry	Trương Hoàng Quí
DS-07	WF-04	Main	Shortlisted application	Schedule → PASS → send gross monthly offer → accept → confirm start	Application PLACED; fee and warranty created	Khúc Ngọc Sơn
DS-08	WF-04	Exception	HEADHUNT_COD application at offer step	Client sends a NET / annual offer	Rejected with HEADHUNT_REQUIRES_GROSS	Khúc Ngọc Sơn
DS-09	WF-05	Main	Placement with warranty set to the shortest configured period	Warranty passes → Admin records fee → approves commission → records payout	Commission PAID; affiliate sees payout	Nguyễn Văn Sang
DS-10	WF-05	Exception	Placement in warranty	Client reports resignation → HR confirms	Warranty VOIDED, commission CANCELLED, replacement slot OPEN	Nguyễn Văn Sang

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	17	3	51
Main workflows (end-to-end)	5	1	5
State transitions of main entities	104	1	104
Agreed NFRs	6	1	6
Committed total			166 (automated in CI ≥ 83)

Automated in CI: at least 50% of the committed total.